

IMMUNOLOGY

Low-dose radiation by radiopharmaceutical therapy enhances GD2 *TRAC*-CAR T cell efficacy in localized neuroblastoma

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Chimeric antigen receptor (CAR) T cells have limited efficacy against solid tumors including neuroblastoma. Here, we evaluated whether low-dose radiation delivered by radiopharmaceutical therapy (RPT), known to potentiate immune checkpoint inhibitors, can synergize with CRISPR-edited GD2 *TRAC*-CAR T cells to improve outcomes in neuroblastoma. We found that in the localized model of neuroblastoma, low-dose radiation delivered by ¹⁷⁷Lu-NM600, an alkylphosphocholine mimetic RPT agent, followed 9 days later by GD2 *TRAC*-CAR T cells led to complete tumor regression. Irradiation of neuroblastoma before GD2 *TRAC*-CAR T cells enhanced the release by CAR T cells of perforin, granzyme B, tumor necrosis factor- α , and interleukin-7 while abrogating transforming growth factor- β 1. Low-dose RPT up-regulated the death receptor Fas on neuroblastoma, potentially enabling CAR-independent killing. This suggests that low-dose RPT can enhance suboptimal CAR T cell efficacy against solid tumors. However, optimization of radiation dose and timing may be needed for each patient and RPT agent to account for varied tumor radiosensitivity and dosimetry.

INTRODUCTION

Chimeric antigen receptor (CAR) T cell therapy has transformed the treatment of hematological malignancies. In patients with treatment-resistant CD19-expressing lymphoma and leukemia as well as B cell maturation antigen-expressing multiple myeloma, CAR T cells have achieved complete remission rate ranging from 54 to 90% (1–4). However, this high efficacy has not been replicated in solid tumors, including neuroblastoma, the most common pediatric extracranial solid malignancy, which expresses high levels of the tumor-associated disialoganglioside GD2 (5–8). This limited efficacy is in part due to limited CAR T cell infiltration into the tumor microenvironment (TME), exhaustion, poor persistence, and antigen escape with little to no antigen expression by the tumor (9).

Development of virus-based third-generation CAR T cells targeting GD2 has recently demonstrated the highest clinical efficacy observed to date in treating relapsed or refractory high-risk neuroblastoma. In a cohort of patients with treatment-resistant neuroblastoma, 33% achieved complete response (CR), 30% had partial response, and 19% showed stable disease (10). However, 3-year event-free survival was modest at 36%, suggesting that more advanced engineering of GD2 CAR T cells may be needed. Promising

strategies such as CRISPR/Cas9 gene editing of CAR transgenes into the T cell receptor α constant (*TRAC*) gene have emerged. *TRAC*-CAR T cells lack endogenous T cell receptors (TCRs) on the cell surface and demonstrate less tonic signaling with enrichment of stem cell memory phenotypes, greater potency, and resistance to exhaustion compared to virally manufactured CAR T cells (11–13).

Radiation promotes immune cell activation and migration into the TME by inducing immunogenic cell death (14, 15). Preclinical evidence suggests a potential synergy between radiation delivered as external beam radiation (EBRT) and CAR T cells against solid tumors. In a model of GD2-expressing glioblastoma, sublethal dose of focal or whole-body irradiation administered by EBRT combined with GD2 CAR T cells led to a durable complete tumor response. In vivo real-time imaging demonstrated that radiation enhanced CAR T cell infiltration and expansion within the TME (16). However, in cases of widespread metastasis where whole-body EBRT may be necessary to target multiple sites of disease to synergize CAR T cell therapy, concerns about acute and long-term toxicities arise. These include lymphopenia, endocrine dysfunctions, cardiovascular diseases, radiation pneumonitis, and infertility (17). Additional risks of secondary malignancies, growth failure, and long-term neurocognitive deficits are particularly worrisome in pediatric patients (18, 19).

Radiopharmaceutical therapy (RPT) encompasses drugs that selectively deliver systemic radiation in vivo via a tumor-selective ligand labeled with a radionuclide with markedly less toxicity than whole-body EBRT (20, 21). Emerging evidence indicates that combining low-dose radiation from RPT with immune checkpoint inhibitors generates a robust antitumor response (22).

Herein, we aim to enhance the therapeutic efficacy of GD2 *TRAC*-CAR T cells against neuroblastoma using a low-dose radiation delivered by lutetium-177 (¹⁷⁷Lu)-based RPT. We found that low-dose RPT enhances the efficacy of GD2 *TRAC*-CAR T cells, and both radiation dose and timing were critical for maximizing the antitumor response.

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Moreover, the enhanced cytotoxicity may be partly mediated by a CAR-independent killing mechanism driven by radiation-induced up-regulation of the death receptor Fas on malignant cells.

RESULTS

Antigen stimulation of GD2 TRAC-CAR T cells during irradiation with ^{177}Lu mitigates the deleterious effect of radiation on CAR T cell viability

GD2 TRAC-CAR T cells were engineered in a nonviral workflow using CRISPR/Cas9 ribonucleoproteins to knock in a third-generation GD2 CAR construct while knocking out the TRAC gene as previously described (12). The construct encoded a GD2-specific single-chain variable fragment 14g2a, a dual costimulatory domain with CD28 and OX40, and a CD3 ζ cytotoxicity domain (fig. S1A). The timeline for GD2 TRAC-CAR T cell production was performed as previously described (fig. S1B) (12). We confirmed proteomic expression of the CAR and TCR knockout in T cells by flow cytometry and confirmed a robust TCR knockout and GD2 CAR knockin (fig. S1C). We previously observed in vitro that radiation delivered directly by ^{177}Lu to GD2 TRAC-CAR T cells resulted in dose-dependent CAR T cell death (23). This suggested that administering GD2 TRAC-CAR T cells after RPT to minimize their exposure to radiation would be the optimal sequence for in vivo studies. Although this sequencing may reduce the irradiation of the GD2 TRAC-CAR T cells, complete avoidance of radiation is not possible. Therefore, we aimed to assess whether antigen stimulation during radiation exposure affects GD2 TRAC-CAR T cell viability. When radiation [2 grays (Gy)] was administered in the presence of GD2-expressing tumors such as CHLA-20 neuroblastoma or M21 melanoma (Fig. 1, A and B, and fig. S2), the viability of GD2 TRAC-CAR T cells is significantly improved. The presence of CHLA-20 and M21 cells led to twofold and 3.5-fold increases in GD2 TRAC-CAR T cell viability, respectively (Fig. 1C; $P < 0.0001$). In contrast, no statistically significant increase in GD2 TRAC-CAR T cell viability was observed in the presence of the GD2-negative CCL-136 rhabdomyosarcoma (Fig. 1, B and C, and fig. S3B). Such antigen stimulation resulted in GD2 TRAC-CAR T cell proliferation, thus mitigating the radiation-induced CAR T cell death (Fig. 1, D and E). This enhanced viability of GD2 TRAC-CAR T cell in the presence of the GD2-expressing cell lines CHLA-20 and M21 was lost when antigen stimulation was delayed until after irradiation (fig. S3C).

Radiation delivered by ^{177}Lu to neuroblastoma cells enhances their vulnerability to GD2 TRAC-CAR T cell-mediated killing and stimulates cytokine secretion

We evaluated whether low-dose radiation by ^{177}Lu sensitizes neuroblastoma cells to killing by GD2 TRAC-CAR T cells. To this end, CHLA-20 cells were irradiated with 2 Gy delivered by ^{177}Lu over 6 days. After irradiation, the viable CHLA-20 cells, determined by a trypan blue assay, were cocultured with GD2 TRAC-CAR T cells at a 10:1 effector-to-target (E:T) ratio for 24 hours (Fig. 2A). Compared to nonirradiated CHLA-20 cells, radiation alone (2 Gy) and GD2 TRAC-CAR T cells alone reduced tumor cell viability by 72.6 and 88%, respectively. However, sequential tumor irradiation followed by GD2 TRAC-CAR T cells resulted in near-complete tumor cell eradication, with a 98% reduction in tumor cell viability (Fig. 2B; $P < 0.0001$). In addition, prior irradiation of CHLA-20 cells led to a significant increase in tumor necrosis factor- α (TNF- α) secretion

by GD2 TRAC-CAR T cells compared to nonirradiated tumor cells (Fig. 2C, left; $P = 0.01$). No statistically significant differences were observed with secretion of other cytokines, including interferon- γ (IFN- γ) (Fig. 2C, right), interleukin-4 (IL-4), or IL-13 (Fig. 2D).

Assessing dosimetry and safety of ^{177}Lu -NM600 in tumor xenograft mouse models

2-(Trimethylammonio) ethyl(18-(4-(2-(4,7,10-tris(carboxymethyl)-1,4,7,10-tetraazacyclododecan-1-yl)acetamido)phenyl)octadecyl)phosphate (NM600; Fig. 3A, top panel) is an alkylphosphocholine (APC) analog with a dodecane tetraacetic acid (DOTA) chelating moiety that selectively delivers radionuclides such as yttrium-90 (^{90}Y) and ^{177}Lu (^{177}Lu -NM600; Fig. 3A, bottom panel) in vivo to a wide range of tumor types, including breast, prostate, lung, and sarcoma (24, 25). Therefore, we investigated whether NM600 can deliver ^{177}Lu in a neuroblastoma xenograft model in NOD-*Rag1*^{null}*IL2 γ* ^{null} (NRG) mice. Longitudinal single-photon emission computed tomography/computed tomography (SPECT/CT) studies were conducted after the intravenous injection of ^{177}Lu -NM600 in NRG mice bearing CHLA-20 tumors (Fig. 3B). Representative maximum-intensity projections of ^{177}Lu -NM600 in vivo at various time points are shown in Fig. 3C, confirming the effective delivery of ^{177}Lu by NM600 to the tumor. An ex vivo biodistribution study was performed 180 hours after the intravenous injection of ^{177}Lu -NM600, revealing the accumulation of ^{177}Lu -NM600 in the tumor and hepatic excretion consistent with a previous report (Fig. 3D) (24). Similarly, NM600 also delivers ^{177}Lu in other xenograft models of GD2-expressing tumors including the melanoma M21 (fig. S4) and the neuroblastoma SY5Y (fig. S5A). Longitudinal SPECT/CT scans were used for dosimetry studies, which estimated an absorbed dose of 0.035 Gy/ μCi for the CHLA-20 tumor. Dosimetry estimates of the absorbed doses for M21 and SY5Y tumors were 0.023 and 0.0219 Gy/ μCi , respectively. Although hepatic excretion of ^{177}Lu -NM600 raises potential concerns for hepatotoxicity, no changes were observed in liver function tests at the highest radiation dose (100 μCi) used for therapeutic studies (Fig. 3E). Moreover, this radiation dose did not adversely affect the differential blood counts or kidney functions (Fig. 3E), demonstrating the safety of low-dose radiation delivered by the activities of ^{177}Lu -NM600 administered in our study. While the radiation dose used herein was safe, it appears to promote angiogenesis as evidenced by the up-regulation of angiogenic markers such as CD34 and CD31 in the TME (Fig. 3F) (26, 27). Furthermore, “cold” NM600 (without ^{177}Lu) had no in vivo effect on tumor growth (fig. S6).

Low-dose radiation delivered by ^{177}Lu -NM600 enhances GD2 TRAC-CAR T cell therapy against a xenograft mouse model of localized neuroblastoma

Given the cooperative efficacy observed in vitro between radiation delivered by ^{177}Lu and GD2 TRAC-CAR T cells, we evaluated this therapeutic effect in a human neuroblastoma flank xenograft model in NRG mice. We evaluated the in vivo efficacy of GD2 TRAC-CAR T cells when administered after two different low doses of radiation, 1.8 and 3.6 Gy. After confirming the subcutaneous engraftment of luciferase-expressing CHLA-20 cells, tumor-bearing mice were randomized into four treatment groups: control [phosphate-buffered saline (PBS)], low-dose radiation alone delivered by ^{177}Lu -NM600, GD2 TRAC-CAR T cells alone, or a combination group receiving low-dose radiation followed by GD2 TRAC-CAR T cell infusion 9 days later (Figs. 4A and 5A). The delay between irradiation and GD2

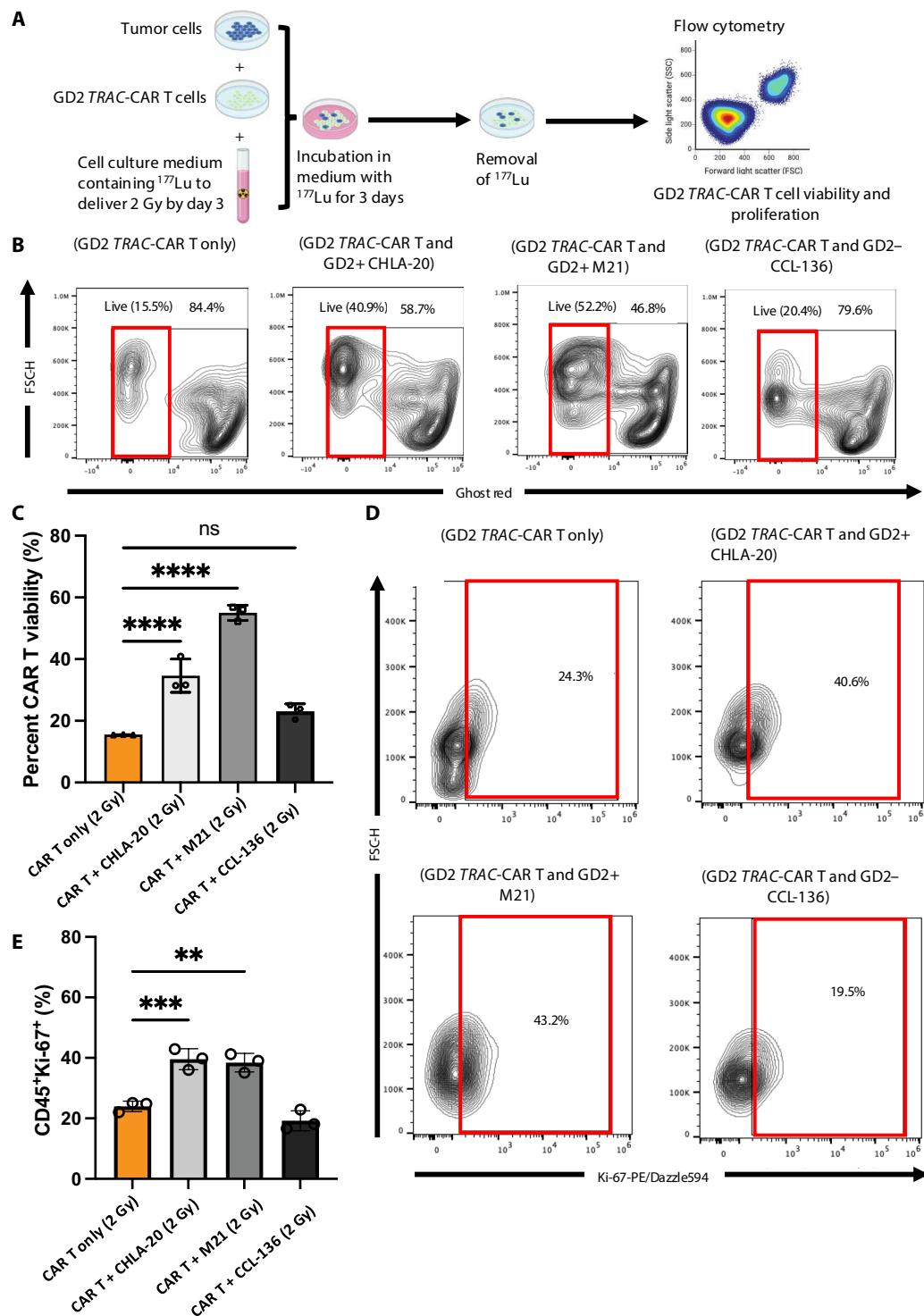


Fig. 1. Antigen stimulation of GD2 TRAC-CAR T cells mitigates the deleterious effect of ^{177}Lu radionuclide exposure on CAR T cell viability. (A) CHLA-20 cells were cocultured with GD2 TRAC-CAR T cells at an E:T ratio of 10:1 in medium containing ^{177}Lu activity capable of delivering 2 Gy to all cells. After 3 days, ^{177}Lu was removed and CAR T cell viability was assessed by flow cytometry. (B) Representative contour plot illustrating the viability of irradiated (2 Gy) GD2 TRAC-CAR T cells (CD45⁺) cocultured with GD2+ tumors (CHLA-20 and M21) and a GD2- tumor (CCL-136). (C) Viability of GD2 TRAC-CAR T cells exposed to ^{177}Lu radionuclide after antigen-dependent versus antigen-independent stimulation. (D) Representative contour plot illustrating the proliferation (Ki-67 expression) of irradiated GD2 TRAC-CAR T cells in the presence of GD2+ or GD2- tumors. (E) Antigen stimulation of GD2 TRAC-CAR T cells during irradiation results in CAR T cell proliferation, evidenced by increased Ki-67 expression. ** $P = 0.0015$; *** $P = 0.0009$; **** $P < 0.0001$. Error bars indicate SD. ns, not significant.

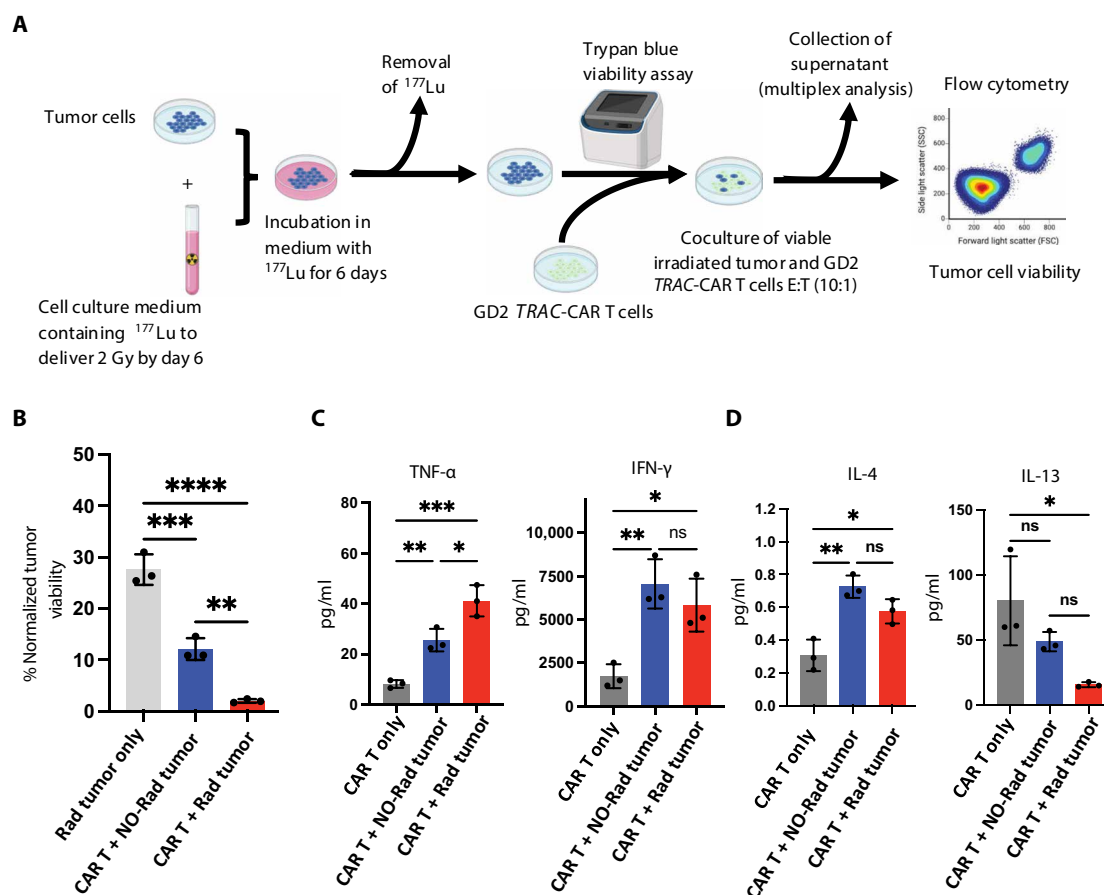


Fig. 2. Radiation delivered by ^{177}Lu to neuroblastoma cells enhances their vulnerability to GD2 TRAC-CAR T cell-mediated killing and stimulates cytokine secretion. (A) CHLA-20 cells were cultured in a medium containing ^{177}Lu , delivering 2 Gy of radiation to tumor cells by day 6. Following radiation delivery, ^{177}Lu was removed and tumor viability was determined using a trypan blue assay. Viable CHLA-20 cells, irradiated and nonirradiated, were then cocultured with GD2 TRAC-CAR T cells at a 10:1 E:T ratio for 24 hours, and tumor viability was measured by flow cytometry. (B) Viability of irradiated CHLA-20 cells after coculture with GD2 TRAC-CAR T cells compared to radiation or TRAC-CAR T cells alone. ** $P = 0.0027$; *** $P = 0.0003$; **** $P < 0.0001$. (C) Secretion of T helper 1 cytokines TNF- α and IFN- γ and (D) T helper 2 cytokines IL-4 and IL-13 by GD2 TRAC-CAR T cells after their coculture with irradiated CHLA-20 cells compared to nonirradiated CHLA-20 cells. TNF- α : * $P = 0.0126$; ** $P = 0.0076$; *** $P = 0.0003$. IFN- γ : * $P = 0.0176$; ** $P = 0.0051$. IL-4: * $P = 0.0150$; ** $P = 0.0017$. IL-13: * $P = 0.0186$. Error bars indicate SD.

TRAC-CAR T cell infusion was intended to minimize TRAC-CAR T cell exposure to radiation, allowing for the decay of ^{177}Lu , which has a half-life of 6.6 days. Two methods—tumor volume and bioluminescence—were used to monitor therapeutic effects.

With a mean tumor dose from RPT of 1.8 Gy, combination RPT + GD2 CAR-TRAC therapy produced a robust treatment response, as compared to all control groups, leading to complete tumor regression in all mice by 29 days after tumor cell implantation (Fig. 4, B and C). No treatment-related toxicity was observed, mice weights remained similar in all four groups (Fig. 4D), and there were no visible signs of xenogeneic graft-versus-host-disease (GVHD) in the groups receiving GD2 TRAC-CAR T cells, consistent with previous reports (12, 28). To assess the durability of response, mice were monitored for up to 125 days after tumor implantation. All mice (eight of eight) in the combination group were alive at day 125 compared to one of five in each monotherapy group and no mice in the untreated control group (Fig. 4E). Among the survivors, all mice receiving combination RPT + GD2 CAR-TRAC achieved a

persistent CR, while only one CR was observed in the low-dose RPT monotherapy group and no CRs were observed in the GD2 TRAC-CAR T cell monotherapy group (Fig. 4F).

Besides the mean tumor dose of 1.8 Gy, we also assessed the efficacy of this therapeutic combination at 3.6 Gy. Our aim was to leverage the tumoricidal effects of a higher radiation dose while maintaining the synergistic benefits with TRAC-GD2 CAR T cells. Initially, the 3.6-Gy mean tumor dose RPT monotherapy group exhibited a therapeutic response similar to that of combination RPT with GD2 TRAC-CAR T cells (Fig. 5, B and C). However, with extended follow-up, combination therapy demonstrated a substantially better outcome compared with RPT monotherapy (Fig. 5E), with five of eight mice receiving the combination therapy showing a persistent CR up to 125 days posttumor implantation versus none of the mice receiving either monotherapy (Fig. 5F). The complete and persistent response rate for the combination of 3.6 Gy RPT and GD2 TRAC-CAR T cells was 62.5% (Fig. 5F) compared to 100% with the combination of 1.8-Gy RPT and GD2 TRAC-CAR T cells (Fig. 4F).

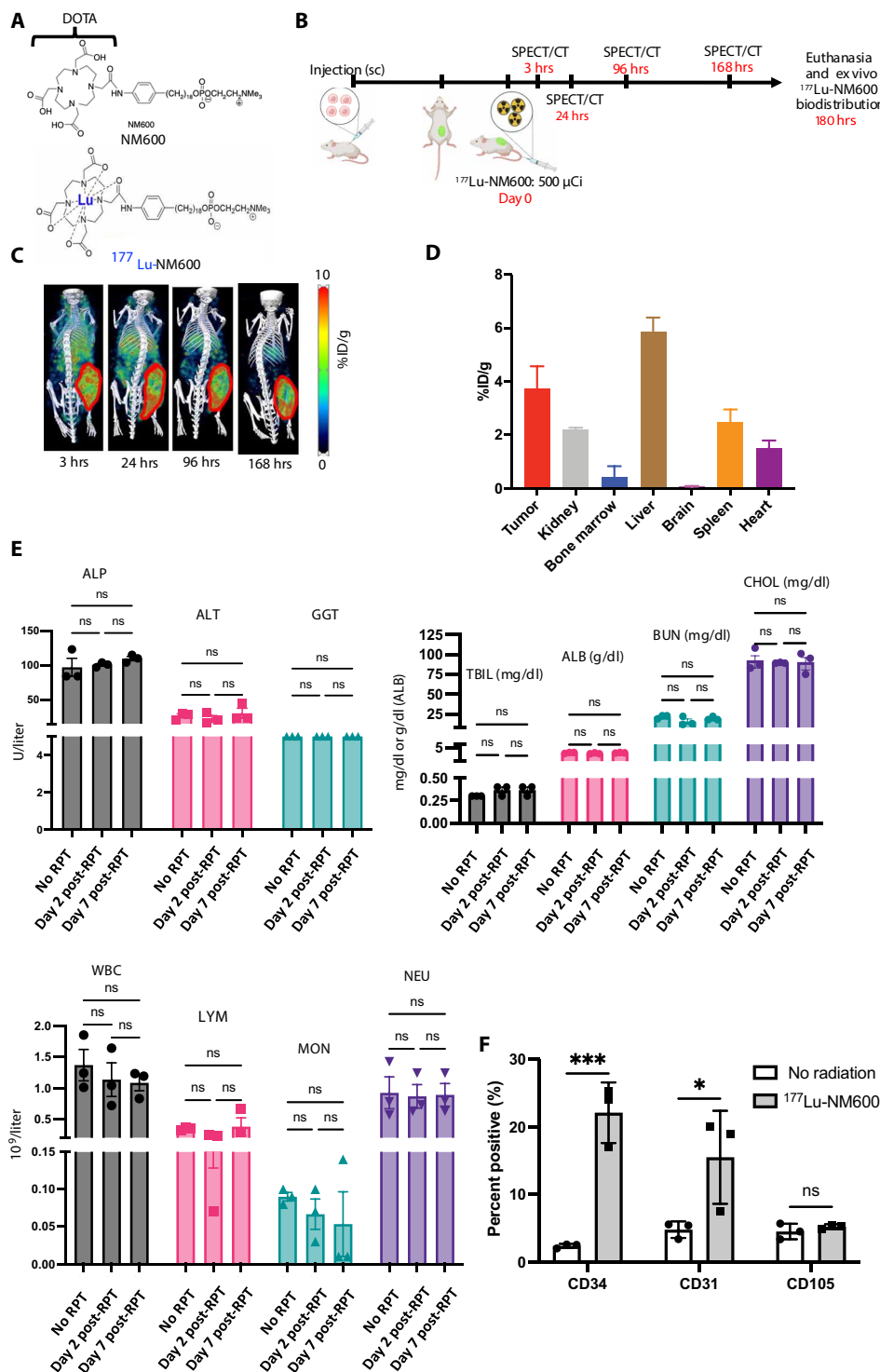


Fig. 3. NM600 RPT delivers ^{177}Lu to a human neuroblastoma xenograft mouse model. (A) Structure of unbound NM600 and NM600 chelating ^{177}Lu (^{177}Lu -NM600). (B) Flank CHLA-20 tumor-bearing NRG mice received ^{177}Lu -NM600 (500 μCi) intravenously, and serial SPECT/CT scans were performed after 3, 24, 96, and 168 hours (hrs). After 180 hours, the mice were euthanized and ex vivo biodistribution of ^{177}Lu -NM600 was evaluated. (C) Representative maximum intensity projection SPECT/CT of ^{177}Lu -NM600 in a CHLA-20 tumor-bearing NRG mouse. sc, subcutaneously. (D) Ex vivo biodistribution of ^{177}Lu -NM600 in the CHLA-20 xenograft model 180 hours after intravenous injection of ^{177}Lu -NM600. (E) The highest radiation dose of 100 μCi used for subsequent in vivo studies is not associated with end-organ toxicities including hepatotoxicity [ALP (alkaline phosphatase), ALT (alanine transaminase), GGT (γ -glutamyl transferase), TBIL (total bilirubin), ALB (albumin), and CHOL (cholesterol)], bone marrow toxicity [WBC (white blood cells), LYM (lymphocytes), MON (monocytes), and NEU (neutrophils)], and nephrotoxicity [BUN (blood urea nitrogen)]. (F) A low dose of ^{177}Lu -NM600 (50 μCi) promotes angiogenesis in the TME. * $P = 0.0355$; *** $P = 0.0002$. Error bars indicate SD.

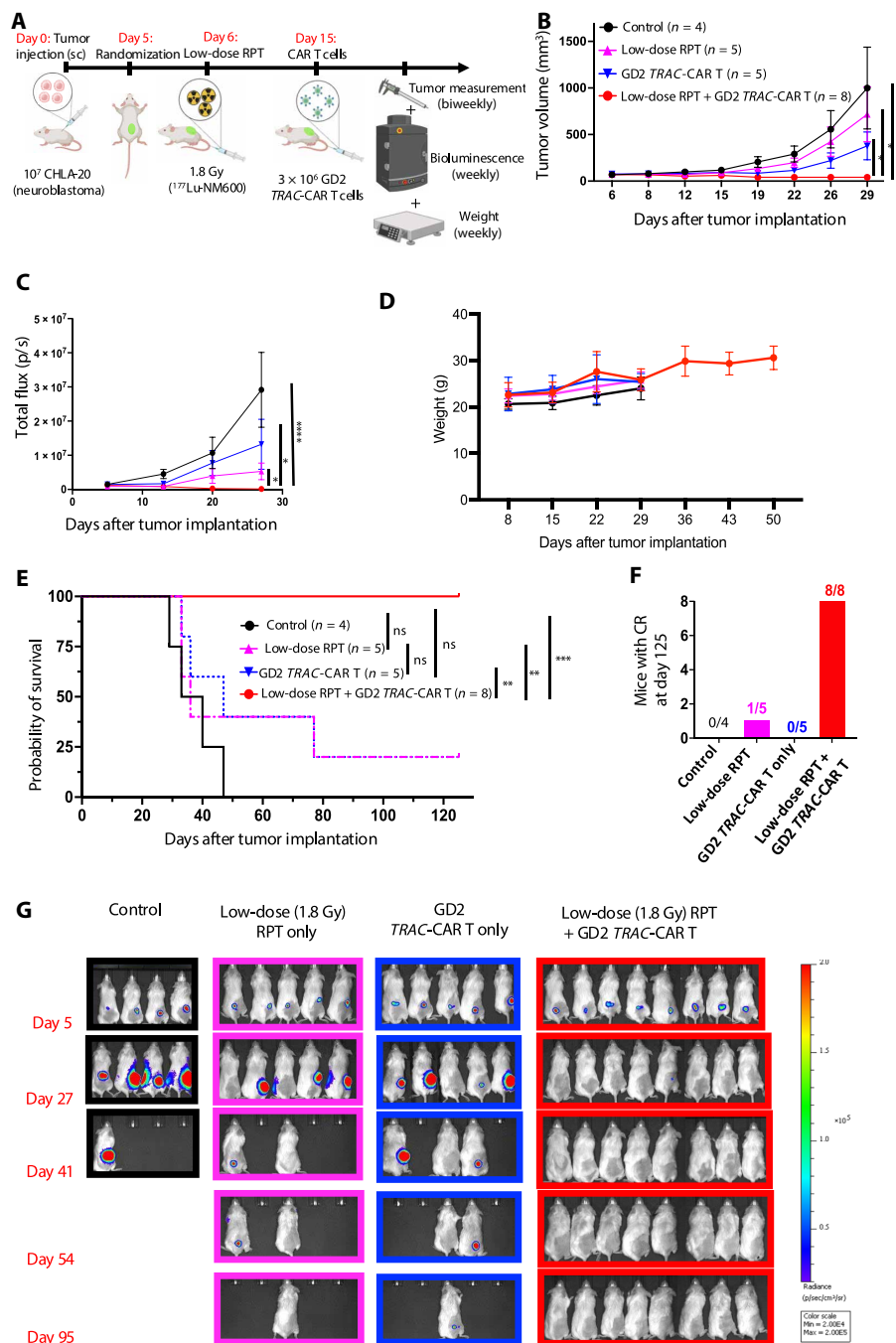


Fig. 4. Low mean tumor dose of 1.8 Gy delivered in vivo by $^{177}\text{Lu-NM600}$ enhances GD2 TRAC-CAR T cell cytotoxicity against a localized human xenograft model of neuroblastoma and improves overall survival. (A) Experimental scheme: 10^7 CHLA-20 cells were subcutaneously injected on the right flank of NRG mice. Tumor implantation was confirmed by IVIS, and mice were randomized into four groups: control (PBS; $n = 4$), low-dose RPT alone (1.8 Gy, $n = 5$), GD2 TRAC-CAR T cells alone (3×10^6 CAR+ cells), and low-dose RPT followed by GD2 TRAC-CAR T cells ($n = 8$) delivered 9 days post-RPT. The treatment effect was assessed by measuring tumor volume and bioluminescence. Treatment-related toxicity was measured by tracking weight loss and behavioral assessments. (B) Tumor volumes between the control group, monotherapy groups, and combination therapy. Complete tumor regression by combination therapy was observed by day 29 posttumor injection (* $P < 0.012$; ** $P = 0.0087$). (C) Total flux (p/s) of tumors as measured by IVIS. The total flux was statistically lower in the combination group than in either monotherapy by day 29 post-tumor injection (two-way ANOVA with Dunnett's multiple comparisons test; * $P < 0.01$; *** $P < 0.0001$). (D) The weight of each mouse was averaged for each group over time. (E) Kaplan-Meier curves: All mice (eight of eight) in the combination group were alive 125 days posttumor injection compared to one of five mice in either monotherapy group (** $P = 0.0019$; *** $P = 0.0002$). (F) CR in each treatment group at day 125 posttumor injection. All mice (eight of eight) had persistent CR in the combination group compared to one of five mice in the low-dose RPT alone group and none in the control and GD2 TRAC-CAR T cell groups. (G) Representative bioluminescence (radiance: p/sec per square centimeter per steradian) images of all treatment groups. The control and GD2 TRAC-CAR T-only group images in this figure are intentionally duplicated (same experimental design).

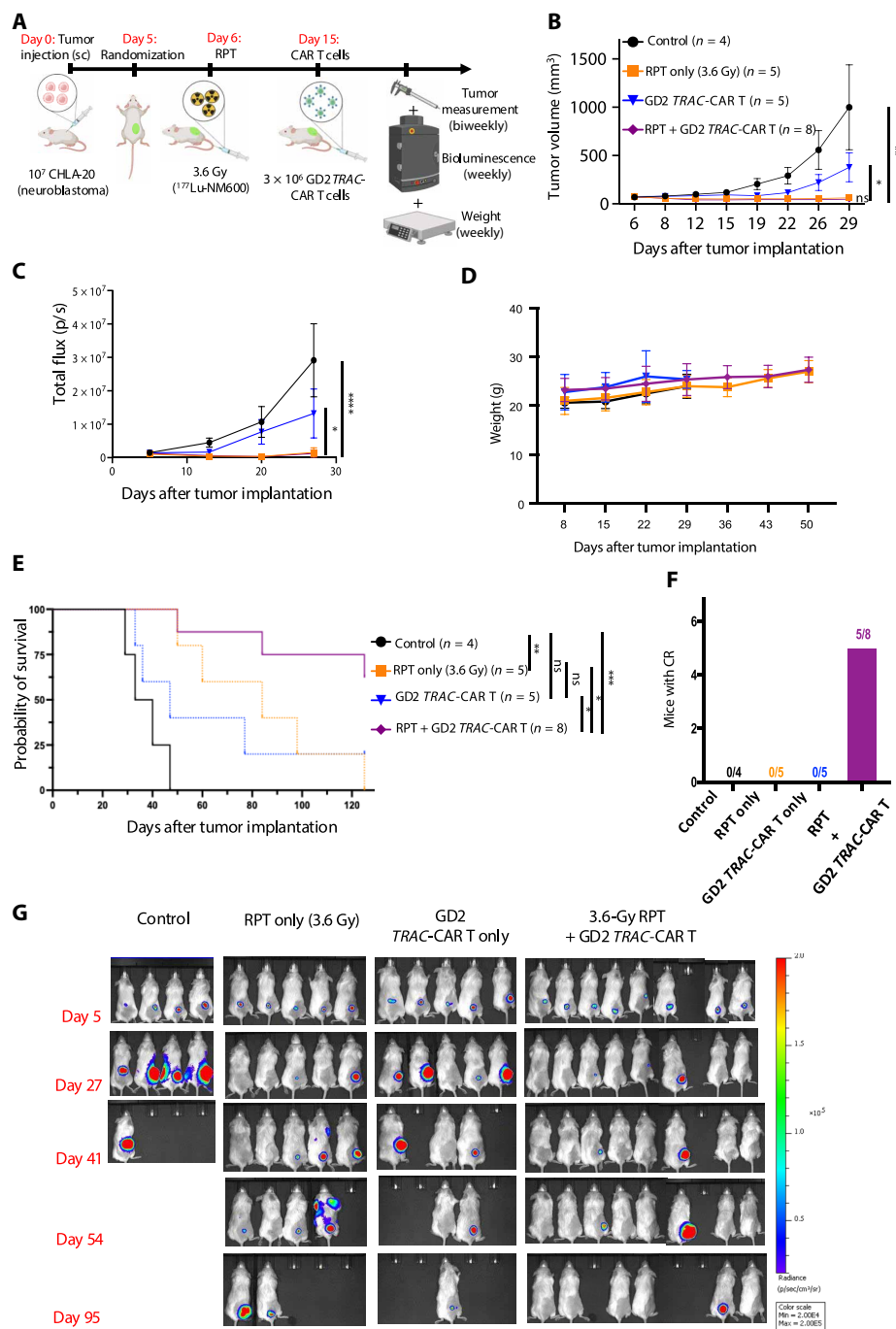


Fig. 5. Low mean tumor dose radiation of 3.6 Gy delivered in vivo by $^{177}\text{Lu-NM600}$ improves the efficacy of GD2 TRAC-CAR T cells against a localized human xenograft model of neuroblastoma. (A) 10^7 CHLA-20 cells were subcutaneously injected on the right flank of NRG mice. Tumor implantation was confirmed by IVIS, and mice were randomized into four groups: control (PBS; $n=4$), 3.6 Gy of RPT alone ($n=5$), GD2 TRAC-CAR T alone (3×10^6 cells), and 3.6 Gy of RPT followed by GD2 TRAC-CAR T cells ($n=8$) delivered 9 days post-RPT. The treatment effect was assessed by measuring tumor volume and bioluminescence. Treatment-related toxicity measured by tracking weight loss and behavioral assessment. (B) Tumor volumes between the control group, monotherapy groups, and combination therapy. Day 29 posttumor injection, the combination led to significant tumor regression compared to the control and GD2 TRAC-CAR T cell alone groups (* $P=0.015$; ** $P=0.009$). (C) The tumor total flux (p/s) was significantly lower in the combination group than in GD2 TRAC-CAR T cell alone and control groups by day 29 posttumor injection (two-way ANOVA with Dunnett's multiple comparisons test; * $P=0.023$; **** $P<0.0001$) but not statistical different from the 3.6-Gy monotherapy group. (D) Weights were averaged in each group over time. (E) Kaplan-Meier curves: 62.5% of mice (five of eight) in the combination group were alive 125 days posttumor injection compared to zero of five mice in either monotherapy group (* $P<0.035$; ** $P=0.0027$; *** $P=0.0002$). (F) CR in each treatment group at day 125 posttumor injection. Five of eight mice had persistent CR in the combination group compared to none in all other groups. (G) Representative bioluminescence (radiance: p/sec per square centimeter per steradian) images of all treatment groups. The control and GD2 TRAC-CAR T-only group images in this figure are intentionally duplicated (same experimental design).

Using the previously performed dosimetry study (Fig. 3), we evaluated the residual tumor activity concentration per injected activity as a function of time ($\mu\text{Ci}/\text{ml}$ per injected microcurie) (fig. S7A). Nine days after RPT delivery, corresponding to the time of GD2 TRAC-CAR T cell infusion, the residual activity concentration in the TME was estimated to be $0.0195 \mu\text{Ci}/\text{ml}$ per injected microcurie. As such, a mean tumor dose of 3.6 Gy, equivalent to an initial administered activity of 100 μCi , yielded a tumor residual activity concentration of $1.95 \mu\text{Ci}/\text{ml}$, twice as much as that of a mean tumor dose of 1.8 Gy (50 μCi), $0.97 \mu\text{Ci}/\text{ml}$. The subsequent exposure of GD2 TRAC-CAR T cells to 1.95 or $0.97 \mu\text{Ci}/\text{ml}$ in vitro resulted in a density of live GD2 TRAC-CAR T cells of 1.16 ± 0.28 million/ml and 1.79 ± 0.074 million/ml, respectively (fig. S7B).

Similarly, the combination of low-dose RPT delivered by ^{177}Lu -NM600 and GD2 TRAC-CAR T cells demonstrated an antitumor activity against a second GD2+ neuroblastoma xenograft model with SY5Y cells (fig. S5). This therapeutic combination also showed

efficacy in targeting another tumor histology, the GD2+ melanoma M21, although with less potency than when tested against neuroblastoma (fig. S8).

In vivo tumor irradiation with 1.8 Gy enhances GD2 TRAC-CAR T cell infiltration in the TME of neuroblastoma

Given that 1.8-Gy mean tumor dose from RPT demonstrated better efficacy when combined with GD2 TRAC-CAR T cells compared to 3.6 Gy, we evaluated the impact of low-dose radiation from ^{177}Lu -NM600 on the infiltration of GD2 TRAC-CAR T cells in the TME. To accomplish this, CHLA-20 tumor-bearing mice were treated with either GD2 TRAC-CAR T cells alone or in combination with low-dose RPT (delivering 1.8-Gy mean tumor dose) administered before CAR T cell infusion, as previously done. Five days after the TRAC-CAR T cell infusion, tumors and spleens were harvested to assess the presence of GD2 TRAC-CAR T cells (Fig. 6A and fig. S9). Our results showed that prior irradiation led to nearly a twofold increase in

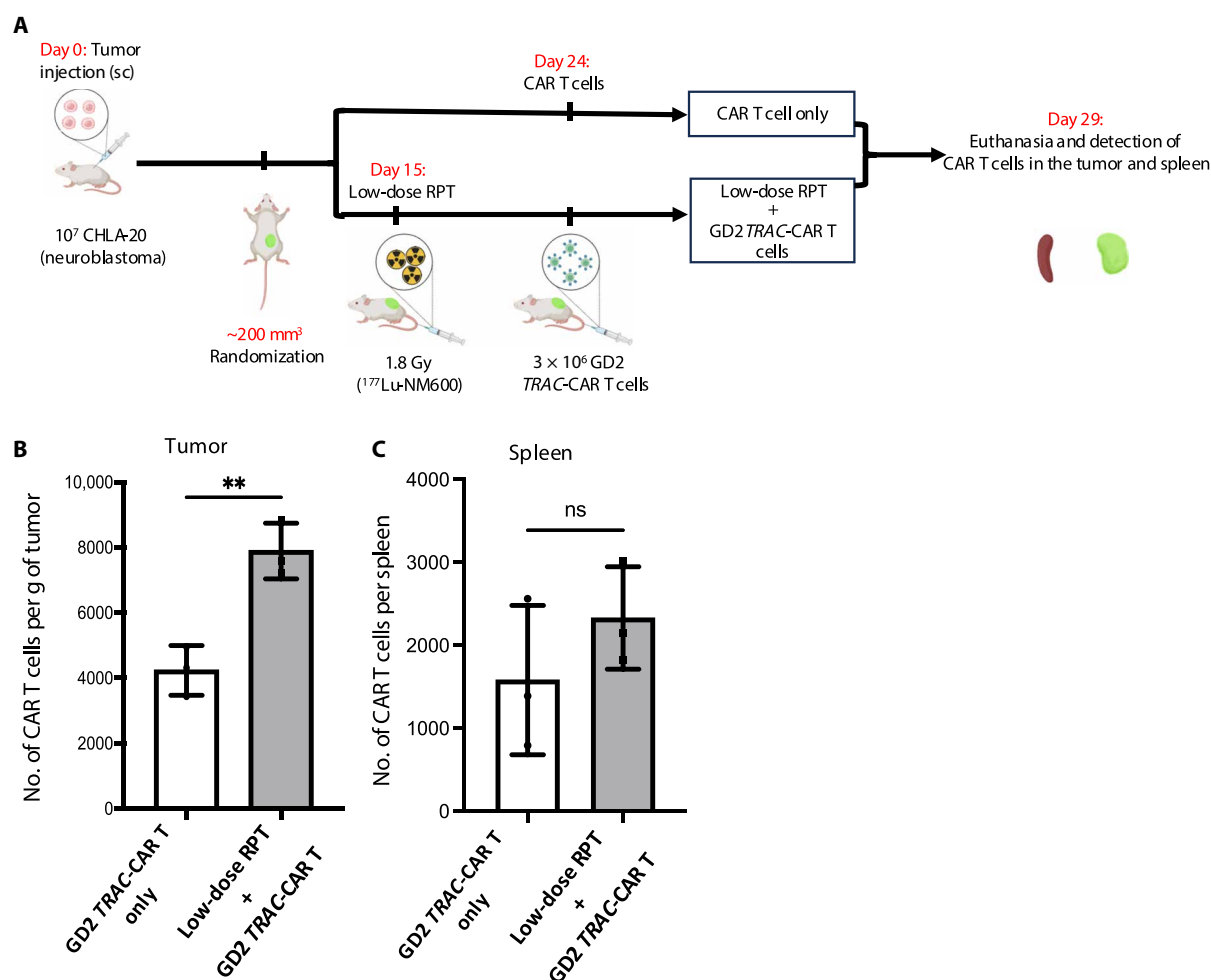


Fig. 6. In vivo tumor irradiation with 1.8-Gy mean tumor dose enhances GD2 TRAC-CAR T cell infiltration in the TME of human neuroblastoma xenograft models. (A) NRG mice bearing large CHLA-20 tumor ($\sim 200 \text{ mm}^3$) were treated either with GD2 TRAC-CAR T cells or with the combination of 1.8-Gy mean tumor dose from RPT followed by CAR T cell infusion delivered 9 days later. Five days after CAR T cell infusion, the mice were euthanized and the number (#) of GD2 TRAC-CAR T cells (B) per gram of tumor and (C) in the whole spleen was analyzed by flow cytometry. A 1.8-Gy mean tumor dose from RPT yielded an approximately twofold increase in infiltration of GD2 TRAC-CAR T cells in the TME compared to administering GD2 TRAC-CAR T cells alone (** $P = 0.0052$). No statistical difference in GD2 TRAC-CAR T cells was noted in the spleen with either treatment.

GD2 TRAC-CAR T cells in the TME (Fig. 6B; $P = 0.0052$). However, no statistically significant difference was observed in the spleens of irradiated versus nonirradiated mice (Fig. 6C).

Irradiation of neuroblastoma enhances the polyfunctionality of CD8⁺ GD2 TRAC-CAR T cells

To assess the functional changes in CD8⁺ GD2 TRAC-CAR T cells induced by radiation from ¹⁷⁷Lu on neuroblastoma, we analyzed the single-cell secretome of CD8⁺ GD2 TRAC-CAR T cells after their coculture (stimulation) with either irradiated (2 Gy) or nonirradiated CHLA-20 neuroblastoma cells using the IsoCode chip (Fig. 7A) (29). Irradiation of neuroblastoma substantially increased the secretion by CD8⁺ GD2 TRAC-CAR T cells of granzyme B (GZMB) and perforin (PRF) while reducing the secretion of inflammatory proteins MCP-1 (monocyte chemoattractant protein-1) and MCP-4 (Fig. 7B). The secretion of chemoattractive cytokines MIP-1 β (macrophage inflammatory protein 1 β) and RANTES was slightly higher with nonirradiated tumor cells (Fig. 7B). The coculturing of CD8⁺ GD2 TRAC-CAR T cells with irradiated tumor cells led to a notable increase in IL-7 secretion by the TRAC-CAR T cells compared to those cocultured with nonirradiated tumor cells. Furthermore, the tumor cell irradiation abolished the secretion of the regulatory/inhibitory cytokine transforming growth factor- β 1 by CD8⁺ GD2 TRAC-CAR T cells (Fig. 7B).

Although the percentage of CD8⁺ GD2 TRAC-CAR T cells secreting a single cytokine was similar under both conditions, tumor cell irradiation led to a substantial increase in the population of CD8⁺ GD2 TRAC-CAR T cells secreting two or more cytokines (Fig. 7C). Notably, CD8⁺ GD2 TRAC-CAR T cells secreting five or more cytokines were detected following tumor irradiation (Fig. 7C). Compared to nonirradiated tumor cells, radiation delivered by ¹⁷⁷Lu resulted in a threefold increase in the overall polyfunctional strength index (PSI) of CD8⁺ GD2 TRAC-CAR T cells. This increase was primarily driven by higher effector and chemoattractive PSIs (Fig. 7D). Moreover, a stimulatory PSI component, absent with nonirradiated tumor, contributed to the overall PSI increase detected after tumor irradiation. Ultimately, the single-cell secretome analysis revealed that the irradiation of neuroblastoma altered the cytokine secretion profile by CD8⁺ GD2 TRAC-CAR T cells, leading to a distinct cluster of CD8⁺ GD2 TRAC-CAR T cells on three-dimensional (3D) uniform manifold approximation and projection (UMAP) compared to nonirradiated neuroblastoma cells (Fig. 7E).

Low-dose radiation from ¹⁷⁷Lu increases the expression of the death receptor Fas on neuroblastoma cells, potentially mediating a CAR-independent killing mechanism by GD2 TRAC-CAR T cells

Radiation delivered by ¹⁷⁷Lu to neuroblastoma cells enhances their sensitivity to GD2 TRAC-CAR T cell killing by boosting two known mechanisms of CAR T cell cytotoxicity: the release of cytokines such as TNF- α (Fig. 2) and cytotoxic effector proteins such as perforin and granzyme (Fig. 7) (30). We evaluated whether this radiation might also impact the death receptor killing pathway mediated by the interaction of Fas on tumor cells with its ligand FasL, a pathway known to trigger CAR-independent killing by CAR T cells (31). To investigate this, CHLA-20 neuroblastoma cells were irradiated with a low dose of radiation (2 Gy), and their expression of Fas was assessed (Fig. 8A and fig. S10). We observed an ~15-fold increase in Fas expression following radiation (Fig. 8B; $P < 0.0001$), along with a significant rise

in median fluorescence intensity (MFI) (Fig. 8C; $P = 0.0005$). In vivo evaluation of the CAR-independent killing mechanism (fig. S11A) revealed that the combination of low-dose radiation (1.8 Gy) and mock (*mCherry*) TRAC-CAR T cells (fig. S11B) was ineffective with disease ultimately recurring in all mice (fig. S11C).

DISCUSSION

Our study demonstrates the preclinical feasibility, safety, and efficacy of combining low-dose radiation delivered by RPT and GD2 TRAC-CAR T cell therapy against localized GD2⁺ solid tumors like neuroblastoma and melanoma. This strategy builds on previous strategies used to enhance CAR T cell therapy against neuroblastoma (10, 12, 32). In addition to enhancing GD2 TRAC-CAR T cell viability, cytotoxicity, and cytokine secretion in vitro, CRs and improved survival with high infiltration in the TME and no GVHD or end-organ toxicity were observed after testing two different low doses of RPT in vivo. Lower doses of RPT like 1.8-Gy mean tumor dose appeared to be more effective than 3.6-Gy mean tumor dose in enhancing survival and CRs by GD2 TRAC-CAR T cells. Single-cell proteomics of CAR T cells showed enhanced secretion of multiple cytokines after exposure to irradiated tumors, especially within effector pathways. Low-dose radiation delivered by RPT may also promote a CAR-independent killing pathway mediated by Fas.

Radiation delivered by EBRT has been shown to potentiate CAR T cell cytotoxicity against various solid tumors (16, 33, 34). For instance, in a xenograft model of non-small cell lung cancer, tumor-targeted low-dose EBRT delivered before mesothelin-targeted CAR T cells enhanced their efficacy (35). This low-dose EBRT promoted chemokine secretion by the tumor, leading to increased infiltration of CAR T cells with up-regulation of corresponding chemokine receptors CXCR3, CXCR6, CCR2, CCR5, and CX3CR1 (35). While low-dose EBRT potentiates CAR T cell therapy against localized solid tumors, its use in metastatic settings may be limited by increased toxicity from larger radiation fields. In such instances, RPT may offer advantages such as the ability to target multiple sites of diseases, including radiographically occult lesions, with potentially less toxicity than EBRT (21). However, RPT poses its own challenges because of the persistence of the radionuclide and ongoing radioactivity decay in vivo, potentially affecting adoptively transferred CAR T cells. A study by Awuah *et al.* demonstrated an improved antitumor activity against multiple myeloma when CS1-CAR T cells were administered 21 days after RPT (²²⁵Ac-DOTA-daratumumab) compared to 14 or 28 days (36). At 14 days post-RPT, residual radiation originating from the remaining RPT can still be detrimental to CAR T cell viability, while at 28 days, an increase in tumor burden can reduce CAR T cell efficacy. This underscores the importance of identifying optimal timing, order, and dose for maximizing antitumor responses from combination RPT and CAR T cell therapy (36).

Yang and colleagues explored the combination of CAR T cells targeting intracellular adhesion molecule 1 or epithelial cellular adhesion molecule with ¹⁷⁷Lu-DOTATATE in xenograft models of gastric cancer (37). In their study, the CAR T cells were administered before RPT, specifically 28 and 16 days before receiving 7.4 MBq (2.6 $\pm$ 1.17 Gy to tumor) or 37 MBq (6 $\pm$ 1.35 Gy to tumor), respectively. Although the combination therapy delayed tumor growth compared to either monotherapy, the sequencing of the therapies implied that CAR T cells within the TME at the time of RPT were exposed to potentially harmful radiation, compromising

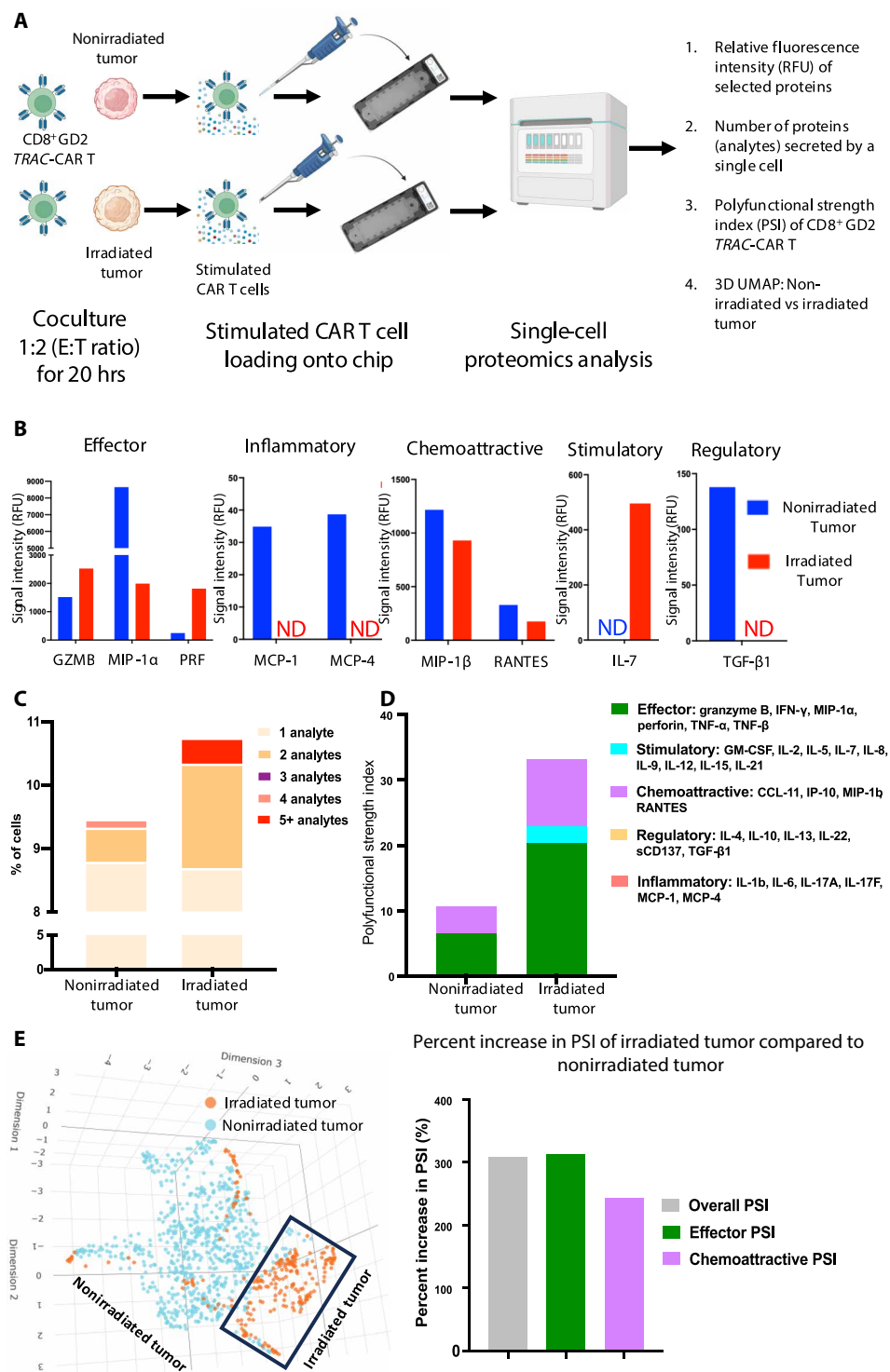


Fig. 7. Irradiation of neuroblastoma induces a polyfunctional phenotype in CD8⁺ GD2 TRAC-CAR T cells. (A) CD8⁺ GD2 TRAC-CAR T cells were cocultured with irradiated or nonirradiated CHLA-20 tumor cells at an E:T ratio of 1:2 for 20 hours. The CD8⁺ GD2 TRAC-CAR T cells were stained and loaded onto an IsoCode chip, and the single-cell secretome was measured for 16 hours. (B) The relative fluorescence intensity (RFU) of selected cytokines/chemokines, categorized as effector, inflammatory, chemoattractive, stimulatory, or regulatory, was measured. (C) Number of individual proteins or analytes secreted by a single CART cell. (D) PSI of CD8⁺ GD2 TRAC-CAR T cells when cocultured with irradiated tumor cells versus nonirradiated tumor cells. (E) 3D UMAP visualization of CD8⁺ GD2 TRAC-CAR T cells under different coculture conditions. CD8⁺ GD2 TRAC-CAR T cells cocultured with irradiated tumor display a distinctive and characteristic proteomic phenotype. ND, not detected.

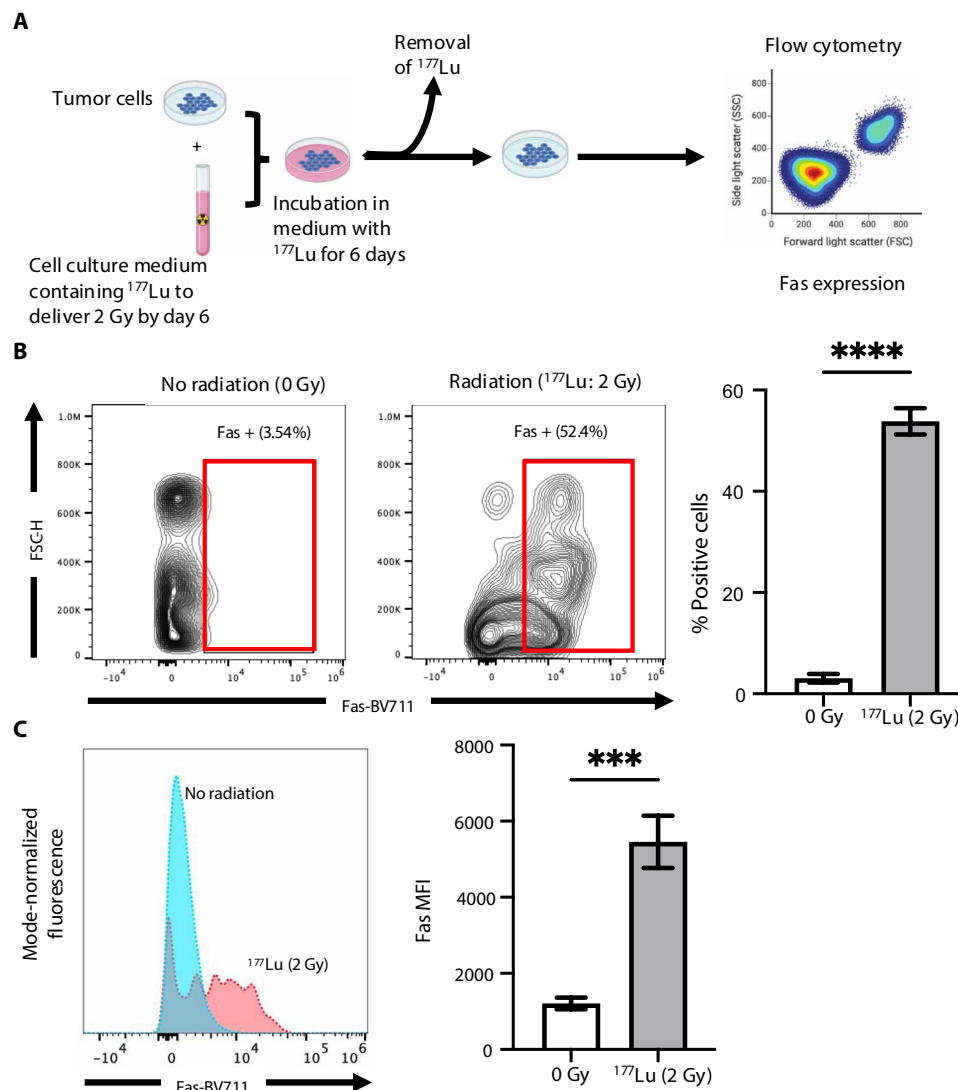


Fig. 8. Irradiation of GD2+ neuroblastoma cells by ^{177}Lu up-regulates Fas expression. (A) CHLA-20 cells were incubated with ^{177}Lu sufficient to deliver 2 Gy by day 6. Fas expression on tumor cells was evaluated by flow cytometry. (B) Fas expression on irradiated neuroblastoma cells was significantly higher than that on nonirradiated neuroblastoma cells. **** $P < 0.0001$. (C) The mean fluorescence intensity (MFI) of Fas on irradiated neuroblastoma cells was also much higher compared to nonirradiated neuroblastoma cells. *** $P = 0.0005$.

their viability and subsequent cytotoxic activity (37). However, the exact mechanism behind the antitumor effects they observed remains unclear.

Minimizing radiation exposure to CAR T cells is paramount to a successful combination regimen. In previous work, we demonstrated that GD2 TRAC-CAR T cells exposed to 1 Gy of radiation delivered by ^{177}Lu retained 87% of their viability in vitro compared to nonirradiated GD2 TRAC-CAR T cells (23). This finding informed the timing of RPT and GD2 TRAC-CAR T cell administration during this current study. Given the 6.6-day half-life of ^{177}Lu , when 1.8-Gy mean tumor dose from RPT (^{177}Lu -NM600) is delivered and followed 9 days later by GD2 TRAC-CAR T cells, the residual radiation dose to the CAR T cells is estimated to be below 1 Gy, suggesting a minimal impact on CAR T cell viability (23). The residual activity concentration in the TME with a mean tumor dose of 1.8 Gy was half that with 3.6 Gy, 0.97 and 1.95 $\mu\text{Ci}/\text{ml}$,

respectively, with the latter residual activity concentration resulting in a significant decrease in the density of live GD2 TRAC-CAR T cell. This may explain why the combination of GD2 TRAC-CAR T cells with 1.8-Gy mean tumor dose was more effective than with 3.6 Gy when RPT was administered 9 days before GD2 TRAC-CAR T cell infusion.

The interaction between the death receptor Fas on tumor cells and its ligand FasL on T cells is a well-known apoptosis-inducing pathway (38, 39). Both EBRT and RPT (^{90}Y -based) have been shown to up-regulate Fas expression on tumor cells, enhancing immune cell-mediated killing (40, 41). Our results suggest that ^{177}Lu similarly up-regulates Fas on neuroblastoma cells, potentially triggering a CAR-independent mechanism for inducing apoptosis of the tumor. This Fas-mediated pathway may be critical for the eradication of antigen-negative tumor cells by T cells and CAR T cells, addressing a barrier of CAR T cell therapy—antigen escape (31).

Single-cell proteomics demonstrated enrichment of secreted effector molecules by GD2 TRAC-CAR T cells when cocultured with irradiated neuroblastoma tumors. Simultaneous up-regulation of perforin and granzyme with TNF- α and IFN- γ production in CD8⁺ GD2 TRAC-CAR T cells cocultured with GD2+ neuroblastoma cells irradiated by ¹⁷⁷Lu-based RPT indicates that this approach maximally activates CAR T cells (42). CAR T cells secreting multiple cytokines have been associated with favorable clinical responses in patients with high-grade lymphoma treated with CAR T cells, suggesting that this therapeutic approach could have potential clinical relevance (43). While our results demonstrated the feasibility of using RPT to enhance suboptimal CAR T cell function and antitumor activity against the solid tumor neuroblastoma, we did observe reduced efficacy with the solid tumor melanoma, potentially due to its inherent radioresistance compared to neuroblastoma (44). While 2 Gy of radiation delivered by ¹⁷⁷Lu was cytotoxic to neuroblastoma cells (CHLA-20) and sensitized them to GD2 TRAC-CAR T cell killing (Fig. 2), such a radiation dose has no such effects on the melanoma cells (M21). Instead, a higher dose (6 Gy) may be needed to sensitize melanoma cells to GD2 TRAC-CAR T cell killing (fig. S12). Thus, caution should be exercised in extrapolating data across all solid tumor histologies.

Several limitations in our study should be noted. The use of an immunodeficient human xenograft mouse model precluded the evaluation of this therapeutic combination in an immunocompetent system, which could have allowed for observations regarding CAR T cells and other lymphocytes like endogenous T, B, and natural killer cells. Considering that low-dose radiation by NM600-based RPT (⁹⁰Y-NM600) has been shown in immunocompetent syngeneic tumor models to enhance the infiltration of endogenous CD8⁺ T cells and natural killer cells in the TME while reducing the infiltration of regulatory T cells, combining low-dose radiation by RPT with CAR T cells could leverage the endogenous immune system to further amplify clinical benefit (22, 45).

In addition, we did not assess the effects of this combination therapy in a widely metastatic or bulky disease model, so it is not yet known whether the degree of tumor burden influences CAR T cell efficacy. Although this therapeutic approach could theoretically be used for all solid tumors, its practical application is constrained to tumors for which both a RPT agent and CAR T cell therapy can be developed, ideally targeting distinct tumor-associated antigens.

In summary, our study demonstrates that low-dose radiation delivered by ¹⁷⁷Lu-based RPT enhances the efficacy of GD2 TRAC-CAR T cells to eradicate GD2+ solid tumors like neuroblastoma, and to a lesser degree, melanoma, by enhancing antigen-dependent and, potentially, CAR-independent mechanisms of cytotoxicity. Complete tumor regression and enhanced overall survival were observed without evidence of end-organ toxicity or xenogeneic GVHD. Factors such as the sequence of administration, radiation dose, heterogeneity in the RPT agent uptake within the TME, radionuclide half-life and linear energy transfer, and timing between RPT and CAR T cell infusion must be carefully considered. Low-dose RPT can enhance otherwise suboptimal CAR T cell function against solid tumors, although additional studies with alternate sources of radiation (α particle emitters or Auger electrons emitters) and CAR T products are required for further optimization in different solid tumor types.

MATERIALS AND METHODS

Cell lines

The human GD2+ tumor cell lines CHLA-20 (neuroblastoma) and M21 (melanoma) were provided by M. Otto (University of Wisconsin-Madison) and R. Reisfeld (Scripps Research Institute), respectively. These cells were cultured in Dulbecco's modified Eagle's medium (DMEM) with high glucose (Gibco), supplemented with 10% fetal bovine serum (FBS) (Avantor) and 1% penicillin-streptomycin (Gibco). GD2+ SY5Y cells (neuroblastoma), donated by C. Mackall (Stanford University), were cultured in RPMI (Gibco) with 10% FBS and 1% penicillin-streptomycin. GD2- CCL-136 cells (rhabdomyosarcoma) were obtained from American Type Culture Collection and grown in DMEM with 10% FBS and 1% penicillin-streptomycin. Tumor cell lines were lentivirally transduced with AkaLUC-GFP (VectorBuilder). All cells were maintained at 37°C in a 5% CO₂ atmosphere. Cell line authentication was performed using genomic short tandem repeat analysis (Idexx BioAnalytics) and morphological assessment according to American Type Culture Collection guidelines. *Mycoplasma* contamination was routinely checked using the Mycoplasma Detection Kit MycoStrip (InvivoGen).

Human T cell isolation and CAR T cell manufacturing

T cells were isolated from leukocyte reduction system cones obtained from the Oklahoma Blood Institute (Oklahoma City, OK) using negative selection with the EasySep Human T Cell Isolation Kit (cat. no. 17951; STEMCELL Technologies). Alternatively, T cells were isolated from peripheral blood of healthy donors under a University of Wisconsin-Madison institutional review board-approved protocol (2017-1070) using the RosetteSep Human T Cell Enrichment Cocktail (cat. nos. 15021 and 15061; STEMCELL Technologies). The sequence for a third-generation GD2-CD28-OX40-CD3 ζ construct was provided by M. Brenner (Baylor College of Medicine). GD2 TRAC-CAR T cells were manufactured by nucleofection of CRISPR/Cas9 into isolated T cells as previously described (12) and cultured in ImmunoCult-XF T Cell Expansion Medium (cat. no. 10981; STEMCELL Technologies) supplemented with IL-7 (10 ng/ml; cat. no. 207-IL-005/CF; BioTechne) and IL-15 (10 ng/ml; cat. no. 247-ILB-005/CF; BioTechne) and maintained at 37°C in 5% CO₂ (13).

In vitro dosimetry

In vitro dosimetry studies were initially conducted in a six-well cell culture plate. The specific activities of ¹⁷⁷Lu delivered as ¹⁷⁷Lu-chloride (¹⁷⁷LuCl₃) required to deliver 2 Gy to a cell monolayer at specific time points were estimated using the Geant4 Monte Carlo toolkit with an extension of RAPID (46). To validate these Monte Carlo predictions, serial dilutions of various ¹⁷⁷Lu activity were prepared in cell culture medium and added to each well. Thermoluminescent dosimeters were placed at the center under each well and analyzed after 10 days by the University of Wisconsin-Madison Radiation Calibration Laboratory (calibration cert. no. 1664.01). A standard curve was generated and used to determine the ¹⁷⁷Lu activity needed to achieve a given radiation dose, as previously described (47). For studies performed in a 96-well plate, a model of a flat-bottom 96-well plate was developed and used in Geant4 Monte Carlo to estimate the specific activities ¹⁷⁷Lu needed to deliver 2 Gy at selected time points.

In vitro GD2 TRAC-CAR T cell irradiation

GD2 TRAC-CAR T cells (10^6 per well on a six-well plate) were irradiated with 2 Gy by ^{177}Lu diluted in 3 ml of ImmunoCult-XF T cell Expansion Medium (cat. no. 10981; STEMCELL Technologies) supplemented with IL-2 (50 IU/ml) over 3 days and then examined by flow cytometry for viability as previously described (23).

In vitro irradiation of GD2 TRAC-CAR T cells in the presence of tumor cells

A total of 100,000 GD2 TRAC-CAR T cells and 10,000 tumor cells was seeded in 100 μl of DMEM in a 96-well plate and irradiated with 2 Gy delivered over 3 days by ^{177}Lu . After 3 days, the cells were harvested and washed three times with PBS. CAR T cell viability was determined by flow cytometry as previously reported (23).

In vitro cytotoxicity activity of GD2 TRAC-CAR T cells against irradiated neuroblastoma cells

A total of 250,000 neuroblastoma cells was irradiated with 2 Gy over 6 days by ^{177}Lu diluted in 3 ml of DMEM in a six-well plate. The cells were harvested and washed three times with PBS. A trypan blue assay was performed to evaluate cell viability, and the viable irradiated neuroblastoma cells were cocultured with GD2 TRAC-CAR T cells at an E:T ratio of 10:1 for 24 hours. The cytotoxic activity of the GD2 TRAC-CAR T cells against irradiated neuroblastoma cells was measured by flow cytometry (23).

In vitro cytokine detection assays

Twenty-four hours after coculturing GD2 TRAC-CAR T cells with irradiated neuroblastoma cells, 50 μl of supernatant was collected and the levels of selected cytokines (TNF- α , IFN- γ , IL-4, and IL-13) were analyzed using the V-PLEX Proinflammatory Panel 1 Kit (Meso Scale Discovery) as previously described (12).

Radiosynthesis of ^{177}Lu -NM600

The radiolabeling of NM600 with $^{177}\text{LuCl}_3$ (SHINE Technologies) was performed as previously reported (24). Briefly, $^{177}\text{LuCl}_3$ was mixed with 55 to 110 nmol of NM600 in 0.5 M NaOAc buffer (pH 5.5). After incubation at 90°C for 30 min, the ^{177}Lu -NM600 complex was purified using reverse-phase chromatography on Sep Pak C18 cartridges (Waters) and eluted in absolute ethanol under N_2 stream. ^{177}Lu -NM600 was reconstituted in saline with 0.4% (v/v) polysorbate 20. The radiosynthesis yield was >95% with a radiochemical purity >98% at an apparent molar activity of 3.4 MBq/nmol.

In vivo dosimetry

In vivo dosimetry of ^{177}Lu -NM600 was estimated as previously reported (24, 46). The total activity in each organ was calculated by extrapolating %IA/g (percent injected activity per gram) at given time points. A standard mouse model was used to convert organ-specific cumulative activity into absorbed dose per injected activity (Gy/MBq). Dose contributions from surrounding organs were included in calculations.

Xenograft mouse model with flank tumor for GD2-TRAC CAR T cell treatment

All procedures were approved by the Institutional Animal Care and Use Committee at the University of Wisconsin-Madison (M005915, M005670, and M006815). Six- to 8-week-old male and female NRG mice (the Jackson Laboratory) were subcutaneously injected with

1×10^7 luciferase-expressing CHLA-20 (AkaLUC-GFP CHLA-20) cells or SY5Y (AkaLUC-GFP SY5Y) or 1.5×10^6 M21 (AkaLUC-GFP M21). Five days after injection, tumor implantation was confirmed by measuring tumor bioluminescence after the intraperitoneal injection of D-luciferin (150 mg/kg) using the In Vivo Imaging System (IVIS) (PerkinElmer). Randomization into the treatment groups was performed on the basis of the bioluminescence imaging signal analyzed with Living Image Software (Revvity) and reported as total flux per second (photon/s). Mice receiving low-dose RPT were administered ^{177}Lu -NM600 on day 6 (posttumor injection), and 3×10^6 (CAR-positive) GD2 TRAC-CAR T cells were administered on day 15. All mice were followed for toxicity, bioluminescence, and overall survival. Disease burden was monitored in vivo through the tumor size and bioluminescence. Images were acquired weekly on the IVIS 15 min after intraperitoneal injection of D-luciferin (150 mg/kg). Tumor dimensions were measured biweekly with an electronic caliper, and volumes were calculated as follows: $(\text{width})^2 \times (\text{length}/2)$. For survival studies, mice were euthanized when the tumor volume reached 1500 mm^3 or tumor ulceration was observed. Weight loss was also monitored for treatment-related toxicity like xenogeneic GVHD. At select time points, tumors and spleens were harvested in FBS-free RPMI medium, mechanically dissociated, and filtered through a 70- μm strainer (Corning). The resulting cell suspensions were centrifuged at 1500 rpm for 5 min, and the pellets were resuspended in ACK lysing buffer (Gibco) for 3 min. After red blood cell lysis, 10 ml of PBS was added, and the samples were centrifuged again at 1500 rpm for 5 min. The pellets were resuspended in PBS and analyzed by flow cytometry.

Flow cytometry analysis

For in vitro analysis, cells were harvested and washed with PBS; samples obtained from in vivo studies were processed as described above. All specimens were resuspended into a single-cell solution in PBS as previously reported (48). Fc blocking with Human TruStain FcX (cat. no. 422302; BioLegend) and Live/Dead staining with Ghost Red Dye 780 (cat. no. 13-0865-T100; Tonbo Biosciences) were performed for 10 min at 4°C. The anti-GD2 scFV on the CAR was identified with 1A7 anti-14G2A antibody (cat. no. Ab02227-10.0; Absolute Antibody) labeled with APC via the APC Conjugation Kit Lightning-Link (cat. no. ab201807; Abcam). Other fluorophore-conjugated antibodies used include anti-GD2-PE-Dazzle594 (cat. no. 357320; BioLegend), anti-CD45-APC (cat. no. 982304; BioLegend), Ki-67-PE/Dazzle594 (cat. no. 350534; BioLegend), anti-CD31-PE/Cyanine7 (cat. no.102418; BioLegend), anti-CD34-PE (cat. no.128610; BioLegend), anti-CD105-AF647 (cat. no.120420; BioLegend), and anti-Fas-BV711 (cat. no. 305644; BioLegend). The fluorophore-conjugated antibodies were incubated for 20 min at 4°C and washed with 2% FBS in PBS. All analyses were performed using the Attune NxT Flow Cytometer (Thermo Fisher Scientific), and data were analyzed with FlowJo version X (BD Biosciences).

Single-cell multiplex secretome analysis

GD2 TRAC-CAR T cells were sorted by flow cytometry to enrich for CD8^+ T cells. The sorted CD8^+ GD2 TRAC-CAR T cells were cocultured with previously irradiated (2 Gy delivered by ^{177}Lu) or nonirradiated CHLA-20 cells at an E:T ratio of 1:2 for 20 hours. Stimulated CD8^+ GD2 TRAC-CAR T cells were then carefully removed, stained with a cell membrane stain 405 (PhenomeX), and loaded onto the IsoCode chip. The single-cell secretome was measured during a

16-hour incubation. Secreted cytokines were identified using a panel of fluorescently labeled antibodies. The data generated were analyzed using IsoPeak software to evaluate the relative fluorescence intensity of the secreted cytokines and the polyfunctional strength and to create 3D UMAP of the CD8⁺ GD2 TRAC-CAR T cells (49).

Statistical analysis

All data generated were analyzed using Prism version 10 (GraphPad). Results are presented as the means $\pm$ SD or standard error of the mean and compared between experimental conditions using two-sample *t* tests or one-way analysis of variance (ANOVA) using Tukey's post hoc test for type I error control (<0.05) when conducting multiple pairwise comparisons. Survival outcomes were analyzed using the Kaplan-Meier method and compared between treatment groups using the log-rank test. All reported *P* values are two-sided, and *P* < 0.05 was used to define statistical significance.

Supplementary Materials

This PDF file includes:

Figs. S1 to S12

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